

Anh Quoc Nguyen

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Center for Theoretical Biological Physics, Rice University - Northeastern University

Homepage

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Google Scholar



RESEARCH INTEREST

Biophysics, Statistical Physics, Tissue Mechanics, Tissue Rheology,
Cell Collective Behavior, Mechanical Memory in Amorphous Materials

EDUCATION

Ph.D. in Biophysics

Sep 2019 – April 2026

Northeastern University, Boston, MA 02115, USA

Thesis: Deciphering Tissue Mechanical Unjamming, Rheology and Plasticity Driven by Cell–Cell Interaction

Bachelor of Science - Physics, Mathematics

Sep 2015–May 2019

Centre College, Danville, KY 40422, USA

GPA: 3.75/4

PUBLICATIONS

- [1] **A. Q. Nguyen**, P. K. Bera, J. Notbohm, and D. Bi, “**Cell-Cell Adhesion as a Double-Edged Sword in Tissue Fluidity**,” *Physical Review E*, 2026, accepted. arXiv: 2603.04170.
- [2] P. K. Bera, **A. Q. Nguyen**, M. McCord, D. Bi, and J. Notbohm, “**Shape-independent fluidity in epithelial cell monolayers**,” *Nature Communications*, 2026, in press.
- [3] **A. Q. Nguyen**, J. Huang, and D. Bi, “**Origin of yield stress and mechanical plasticity in model biological tissues**,” *Nature Communications*, vol. 16, no. 1, p. 3260, 2025.
- [4] S. Kayal, **A. Q. Nguyen**, and D. Bi, “**The Rheology of Living Tissues: From Cells to Organismal Mechanics**,” *Annual Review of Condensed Matter Physics*, vol. 17, pp. 285–304, 2025.
- [5] H. Yang, G. Roy, **A. Q. Nguyen**, D. Bi, T. Stern, M. J. Buehler, and M. Guo, “**MultiCell: geometric learning in multicellular development**,” *Nature Methods*, vol. 23, pp. 617–625, 2026.
- [6] G. Cai, **A. Q. Nguyen**, Y. Bashirzadeh, S.-S. Lin, D. Bi, and A. P. Liu, “**Compressive stress drives adhesion-dependent unjamming transitions in breast cancer cell migration**,” *Frontiers in Cell and Developmental Biology*, vol. 10, 933042, 2022.

PRESS

- “Living tissues may form like avalanches, Northeastern researchers say — a discovery that could aid new treatments”
Northeastern Global News; republished by *Phys.org* Apr 2025

SELECTED CONFERENCES AND ACADEMIC PROGRAMS

- APS Global Summit 2025 (**Oral**) Anaheim - Mar 2025
- APS March Meeting 2024 (**Oral**) Minneapolis - March 2024
- International Physics of Living Systems Meeting (**Oral & Poster**) Atlanta - Aug 2023
- GRC: Cell Contact and Adhesion (**Poster**) New Hampshire - June 2023
- APS March Meeting 2023 (**Oral**) Las Vegas - March 2023

- Summer School for Soft Solids and Complex Fluids (**Poster**)
- APS Division of Fluid Dynamics Meeting (**Oral & Poster**)

UMass Amherst - June 2022
Atlanta - Nov 2018

TEACHING EXPERIENCE

- **Lecturer:** Lab for Physics for Life Sciences 1, Lab for Physics for Life Sciences 2
- **Teaching Assistant:** Physics for Life Sciences 1, Electricity and Magnetism 2, Modern Physics, Computational Problem Solving in Physics, Lab for Physics for Life Sciences 1 & 2, Lab for Physics for Engineering 1 & 2
- **Tutor:** Calculus 1, 2, 3; Linear Algebra; Differential Equations; Classical Mechanics; Probability and Statistics

SKILLS

- **MATLAB:** developed vertex and Voronoi model simulations of confluent tissue, including energy minimization, quasi-static and steady-shear deformation protocols, and soft glassy rheology calculations; analyzed simulation and experimental data
- **Python:** analyzed simulation and experimental data; applied the fastDDM package for differential dynamic microscopy analysis
- **Other:** LaTeX